Requirements Gathering

Intelligent Payment Fraud Detection and Risk-Based Decision System

Data Science and Artificial Intelligence Track

3 Requirements Gathering

3.1 Stakeholder Identification

The project is designed to support users involved in financial transaction monitoring and fraud investigation. The main stakeholders include:

- **Financial Institutions:** Banks and payment service providers interested in reducing fraud-related losses.
- **Fraud Analysts:** Professionals responsible for reviewing suspicious transactions and investigating potential fraud.
- **Risk Management Teams:** Teams that assess financial risks and establish transaction review policies.
- **System Administrators:** Users responsible for managing system access, data handling, and system operation.

3.2 User Needs

The identified stakeholders require a system that can identify potentially fraudulent transactions, prioritize suspicious cases, and provide understandable explanations for its predictions.

The system should also help users understand the trade-off between detecting fraudulent transactions and incorrectly flagging legitimate ones. Results should be presented through a clear dashboard that supports analysis and informed decision-making.

3.3 Functional Requirements

The system shall provide the following functionalities:

1. **Data Input:** Load transaction data from supported files, such as CSV.
2. **Data Preprocessing:** Handle missing values, prepare features, and validate input data before analysis.
3. **Fraud Detection:** Use a trained Machine Learning model to estimate the likelihood that a transaction is fraudulent.
4. **Risk Scoring:** Assign a risk score and categorize transactions into risk levels according to configurable thresholds.
5. **Prediction Explainability:** Present the main features influencing selected predictions using an appropriate explainability method.

6. **Dashboard Visualization:** Display transaction statistics, risk distributions, suspicious transactions, and model performance metrics.
7. **Model Evaluation:** Report relevant metrics, including Precision, Recall, F1-score, and PR-AUC.
8. **Decision Support:** Allow users to explore how different risk thresholds affect fraud detection, false alarms, and missed fraudulent transactions.

3.4 Non-Functional Requirements

- **Performance:** Process supported datasets and generate predictions within a reasonable time using the available hardware.
- **Reliability:** Evaluate the model on unseen data and maintain consistent processing procedures.
- **Security and Privacy:** Restrict access to sensitive data, avoid exposing unnecessary transaction information, and handle files securely.
- **Usability:** Provide a clear, intuitive dashboard with understandable labels, visualizations, and results.
- **Maintainability:** Organize the source code into modular components to simplify testing and future updates.
- **Scalability:** Design the data-processing and prediction pipeline so that it can be extended to larger datasets when resources permit.
- **Interpretability:** Present prediction explanations in a form that users can understand without implying that the model proves fraudulent intent.

3.5 Data Requirements

The system will primarily use the IEEE-CIS Fraud Detection dataset available on Kaggle.

The required data includes transaction-related features and the corresponding fraud labels available for model training and evaluation. The dataset must be inspected to identify missing values, categorical variables, class imbalance, and potential data leakage.

Data preprocessing and feature selection must be performed carefully. Training and evaluation procedures must prevent information from the validation or test data from leaking into model training.

Because the dataset is not specific to Egypt, its results will not be treated as evidence of equivalent performance on Egyptian banking transactions.

3.6 Technical Requirements

The project will use Python as the primary programming language, supported by suitable Data Science and Machine Learning libraries, such as Pandas, NumPy, and Scikit-learn.

Additional tools may include visualization libraries, an appropriate explainability library, and a framework such as Streamlit for the interactive dashboard. Git and GitHub will be used for version control and collaborative development.

The final technology stack will depend on the team's implementation choices and available computational resources.

3.7 Constraints and Assumptions

- The project depends on the availability and suitability of the public IEEE-CIS dataset.
- Development is limited by the team's schedule, skills, and computational resources.
- Model predictions depend on the quality and representativeness of the available data.
- The system will operate as an academic prototype and will not directly authorize, reject, or block real financial transactions.
- Risk scores and explanations will support investigation rather than provide definitive proof of fraud.

3.8 Acceptance Criteria

The project will be considered functionally complete when:

1. The system successfully loads and preprocesses supported transaction data.
2. A trained model generates fraud predictions and risk scores.
3. Model performance is evaluated using appropriate metrics on unseen data.
4. The dashboard displays predictions, risk levels, and relevant evaluation results.
5. An explainability method provides feature-based insights for selected predictions.
6. The system demonstrates how decision thresholds affect false positives and missed fraud cases.
7. The source code, testing results, and technical limitations are documented in the team's GitHub repository.

3.9 Conclusion

These requirements define the expected functionality, quality attributes, data needs, and technical constraints of the Intelligent Payment Fraud Detection and Risk-Based Decision System. They will guide system development and provide a clear basis for testing whether the final prototype meets its intended objectives.